



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

 Website: www.ktunotes.in

CST476 MOBILE COMPUTING

Jo Cheriyan¹

¹*Department of Computer Science and Engineering*

Mobile Security and Next Generation Networks

Security issues in mobile computing

Mobile computing provides a variety of wireless devices that has the mobility to allow people to connect to the internet. It provides wireless transmission to access data and information from the locations they are stored.

There are mainly three aspects of Mobile computing

- Mobile communication: This aspect specifies the communication issues in ad-hoc, infrastructure networks, communication properties, protocols, data formats and concrete technologies.
- Mobile hardware: This aspect specifies the mobile devices or device components that are used in mobile computing.
- Mobile software: This aspect specifies all the necessary files and software related to the computer used in mobile computing.

Mobile Computing faces many improper and unethical practices and problems such as hacking, industrial espionage,

pirating, online fraud and malicious destruction etc. The threats and security issues of mobile computing can be divided into two categories:

- The security issues that are related to the transmission of information over wireless networks.
- The security issues and threats that are related to information and data residing on mobile devices.

□ General Security Issues

There are mainly five fundamental goals of security used in the information system to deal with security issues.

They are:

- **Confidentiality:** This is used to prevent unauthorized users from gaining access to any particular user's critical and confidential information.
- **Integrity:** This is used to ensure that any type of unauthorized modification, destruction or creation of information cannot be done.
- **Availability:** The availability is used to ensure that authorized users get the required access whenever they need it.
- **Legitimate:** This is used to ensure that only authorized, and legitimate users have access to the services.

- **Accountability:** Accountability is used to ensure that the users will be responsible for their security-related activities by arranging the users and their activities in a linked form.

We have to achieve these goals according to the security policy used by the service providers.

□ Wireless Security Issues

Wireless security issues are considered as the primary security issues of mobile computing. These are related to wireless networks. These issues occur when the hackers intercept the radio signals. Most wireless networks are dependent on other private networks, which are managed by others, so after these issues, the users have less control of security procedures.

These security issues are:

- **Denial of Service (DOS) attacks:** The denial of services or DOS attacks is one of the most common attacks of all kinds of networks and especially in a wireless network. It prevents users from using network services because the attacker sends a large amount of unnecessary data or connection requests to the communication server. It causes a slow network, and therefore the users cannot get benefited from using its service.

- **Traffic Analysis:** Traffic analysis is used to identify and monitor communication between users. In this process,

the service provider listens the traffic flowing in the wireless channel to access the private information of users affected by the attacker.

- **Eavesdropping:** It specifies that the attacker can log on to the wireless network and access sensitive data if the wireless network was not secure enough. This can also be done if the information is not encrypted.
- **Session Interception and Messages Modification:** It specifies that the attacker can intercept the session and modify the transmitted data in this session. This scenario is called "man in the middle." It inserts the attacker's host between the sender and receiver host.
- **Spoofing:** In this security issue, the attacker impersonates him as an authorized account of another user and tries to access the sensitive data and unauthorized services.
- **Captured and Re-transmitted Messages:** In this security issue, the attacker can get some of the network services by getting unauthorized access. After capturing the message, he/she can reply to it with some modifications to the same destination or another.

□ How to handle security issues?

The biggest issue in mobile computing is the credential verification of users. Because the users share the username and passwords, it may become a significant threat to security. Some recommendations can be followed by companies or mobile users to keep their mobile devices and the data stored in the devices secure.

- You should install security hardware and software.
- You should ensure that the data stored in the mobile devices are encrypted and audited.
- Educate the users on proper mobile computing ethics and security issues.
- Ensure that the mobile devices are configured with a power-on authentication to prevent unauthorized access.
- Ensure that anti-virus software is installed on mobile devices.
- Make sure that the firewall client is installed on mobile devices.
- Make your mobile devices encrypted with a strong password.
- Encrypt your data stored in the secondary storage devices such as Memory Sticks, Data card, removable USB etc.
- Ensure that the Bluetooth, Wi-Fi, etc. enabled mobile devices are turned off when you are not using them.
- Make periodic backups of your mobile devices on a data server.

□ Information security

Information security is the practice of protecting information by mitigating information risks. It involves the protection of information systems and the information processed, stored and transmitted by these systems from unauthorized access, use, disclosure, disruption, modification or destruction. This includes the protection of personal information, financial information, and sensitive or confidential information stored in both digital and physical forms. Effective information security requires a comprehensive and multi-disciplinary approach, involving people, processes, and technology.

Information Security is not only about securing information from unauthorized access. Information Security is basically the practice of preventing unauthorized access, use, disclosure, disruption, modification, inspection, recording or destruction of information. Information can be a physical or electronic one. Information can be anything like Your details or we can say your profile on social media, your data on mobile phone, your biometrics etc. Thus Information Security spans so many research areas like Cryptography, Mobile Computing, Cyber Forensics, Online Social Media, etc.

Effective information security requires a comprehensive approach that considers all aspects of the information

environment, including technology, policies and procedures, and people. It also requires ongoing monitoring, assessment, and adaptation to address emerging threats and vulnerabilities.

Why we use Information Security?

We use information security to protect valuable information assets from a wide range of threats, including theft, espionage, and cybercrime. Information security is necessary to ensure the confidentiality, integrity, and availability of information, whether it is stored digitally or in other forms such as paper documents. Here are some key reasons why information security is important:

- **Protecting sensitive information:** Information security helps protect sensitive information from being accessed, disclosed, or modified by unauthorized individuals. This includes personal information, financial data, and trade secrets, as well as confidential government and military information.
- **Mitigating risk:** By implementing information security measures, organizations can mitigate the risks associated with cyber threats and other security incidents. This includes minimizing the risk of data breaches, denial-of-service attacks, and other malicious activities.
- **Compliance with regulations:** Many industries and jurisdictions have specific regulations governing the protection

of sensitive information. Information security measures help ensure compliance with these regulations, reducing the risk of fines and legal liability.

- Protecting reputation: Security breaches can damage an organization's reputation and lead to lost business.

Effective information security can help protect an organization's reputation by minimizing the risk of security incidents.

- Ensuring business continuity: Information security helps ensure that critical business functions can continue even in the event of a security incident. This includes maintaining access to key systems and data, and minimizing the impact of any disruptions.

Information Security programs are built around 3 objectives, commonly known as CIA - Confidentiality, Integrity, Availability.

- Confidentiality - means information is not disclosed to unauthorized individuals, entities and process. For example if we say I have a password for my Gmail account but someone saw while I was doing a login into Gmail account. In that case my password has been compromised and Confidentiality has been breached.

- Integrity - means maintaining accuracy and completeness of data. This means data cannot be edited in an unauthorized way. For example if an employee leaves an organisation then in that case data for that employee in all

departments like accounts, should be updated to reflect status to JOB LEFT so that data is complete and accurate and in addition to this only authorized person should be allowed to edit employee data.

- Availability - means information must be available when needed. For example if one needs to access information of a particular employee to check whether employee has outstanding the number of leaves, in that case it requires collaboration from different organizational teams like network operations, development operations, incident response and policy/change management.

Denial of service attack is one of the factor that can hamper the availability of information.

□ Information security - Algorithms

Encryption is a way for data-messages or files-to be made unreadable, ensuring that only an authorized person can access that data. Encryption uses complex algorithms to scramble data and decrypts the same data using a key provided by the message sender. Encryption ensures that information stays private and confidential, whether it's being stored or in transit. Any unauthorized access to the data will only see a chaotic array of bytes.

Here are some essential encryption terms you should know:

Algorithm: Also known as a cipher, algorithms are the rules or instructions for the encryption process. The key length, functionality, and features of the encryption system in use determine the effectiveness of the encryption.

Decryption: Decryption is the process of converting unreadable ciphertext to readable information.

Key: An encryption key is a randomized string of bits used to encrypt and decrypt data. Each key is unique, and longer keys are harder to break. Typical key lengths are 128 and 256 bits for private keys and 2048 for public keys. There are two kinds of cryptographic key systems, symmetric, and asymmetric.

Common Encryption Algorithms

1. Triple DES

Triple DES was designed to replace the original Data Encryption Standard (DES) algorithm, which hackers eventually learned to defeat with relative ease. At one time, Triple DES was the recommended standard and the most widely used symmetric algorithm in the industry.

Triple DES uses three individual keys with 56 bits each. The total key length adds up to 168 bits, but experts would argue that 112-bits in key strength is more accurate. Despite slowly being phased out, Triple DES has, for the most part, been replaced by the Advanced Encryption Standard (AES).

2. AES

The Advanced Encryption Standard (AES) is the algorithm trusted as the standard by the U.S. Government and numerous organizations. Although it is highly efficient in 128-bit form, AES also uses keys of 192 and 256 bits for heavy-duty encryption purposes.

AES is largely considered impervious to all attacks, except for brute force, which attempts to decipher messages using all possible combinations in the 128, 192, or 256-bit cipher.

3. RSA Security

RSA is a public-key encryption algorithm and the standard for encrypting data sent over the internet. It also happens to be one of the methods used in PGP and GPG programs. Unlike Triple DES, RSA is considered an asymmetric algorithm due to its use of a pair of keys. You've got your public key to encrypt the message and a private key to decrypt it. The result of RSA encryption is a huge batch of mumbo jumbo that takes attackers a lot of time and processing power to break.

4. Blowfish

Blowfish is yet another algorithm designed to replace DES. This symmetric cipher splits messages into blocks of 64

bits and encrypts them individually. Blowfish is known for its tremendous speed and overall effectiveness.

Meanwhile, vendors have taken full advantage of its free availability in the public domain. You'll find Blowfish in software categories ranging from e-commerce platforms for securing payments to password management tools, where it protects passwords. It's one of the more flexible encryption methods available.

5. Twofish

Computer security expert Bruce Schneier is the mastermind behind Blowfish and its successor Twofish. Keys used in this algorithm may be up to 256 bits in length, and as a symmetric technique, you only need one key. Twofish is one of the fastest of its kind and ideal for use in hardware and software environments. Like Blowfish, Twofish is freely available to anyone who wants to use it.

□ Next Generation Network (NGN)

Next Generation Network (NGN) refers to a packet-based network and it can be used for both telecommunication services as well as data and it supports mobility. It is able to make use of multiple broadband capabilities, especially Quality of Services (QoS) enabled transport technologies where the service-related functions are independent of the underlying transport-related technologies.

The main goal of Next Generation Network (NGN) is to serve/work as an replacement of Public Switched Telephone Network (PSTN) and Integrated Services Digital Network (ISDN).

Nowadays there exist different networks for different purposes like voice, data, and all sorts of video. The telephony, Internet, and cellular networks refer to different domains and each has its own protocol and services. The idea behind the Next Generation Network (NGN) is that one network which transports all type of data and provides services in the form of packets similar to those which is used on Internet. Commonly NGNs are built around Internet Protocol (IP). That's why sometimes NGN is referred to as an all-IP network to describe the transformation towards Next Generation Network (NGN).

International Telecommunication Union Telecommunication Standardization Sector (ITU-T) defines Next

Generation Network (NGN) is a packet based network which provides telecommunication services to users and it uses multiple broadbands, QoS enabled transport technologies. NGN is a generalized mobility approach and it allows consistent and all over-provision of services to users.

Fundamental characteristics for defining NGN:

- Packet based transfer.
- Separation of control functions for bearer capabilities, call/session - service.
- Decoupling of service and network, and provision of open interfaces (I/F).
- A wide range of services like (Real Time(RT)/ streaming/ non Real Time (RT)/ multimedia).
- Interworking with legacy networks via open I/F.
- Support of Generalized mobility.
- Support to unrestricted access by users to different service providers.

Services supported in NGN:

- Data communication services.
- Multimedia Services.

- Public interest services.
- Public Switched Telephone Network (PSTN)/Integrated Services Digital Network (ISDN) simulation services.
- Public Switched Telephone Network (PSTN)/ Integrated Services Digital Network (ISDN) emulation services

Next Generation Network Layer:

- In Access Layer, different types of media gateways that support connection to and from access network with the core network are included.
- Core network layer is network handling converged services based on Internet Protocol (IP). Control layer works as call server. It provides call control functions also provides control of a media gateway.
- Service layer is an IT platform that creates a service creation environment extending its functionality in order to cover new network scenarios as an intelligent network.

□ Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal frequency-division multiplexing is a method of data transmission where a single information stream is split among several closely spaced narrowband subchannel frequencies instead of a single Wideband channel frequency. It is mostly used in wireless data transmission but may be employed in wired and fiber optic communication as well.

In a traditional single-channel modulation scheme, each data bit is sent serially or sequentially one after another. In OFDM, several bits can be sent in parallel, or at the same time, in separate substream channels. This enables each substream's data rate to be lower than would be required by a single stream of similar bandwidth. This makes the system less susceptible to interference and enables more efficient data bandwidth.

How orthogonal frequency-division multiplexing works

In the traditional stream, each bit might be represented by a 1 nanosecond segment of the signal, with 0.25 ns spacing between bits, for example. Using OFDM to split the signal across four component streams lets each bit be represented by 4 ns of the signal with 1 ns spacing between. The overall data rate is the same, 4 bits every 5 ns, but the signal integrity is higher.

OFDM builds on simpler frequency-division multiplexing (FDM). In FDM, the total data stream is divided into several subchannels, but the frequencies of the subchannels are spaced farther apart so they do not overlap or interfere. With OFDM, the subchannel frequencies are close together and overlapping but are still orthogonal, or separate, in that they are carefully chosen and modulated so that the interference between the subchannels is canceled out.

OFDM advantages and disadvantages

Orthogonal frequency-division multiplexing has many advantages over a single-channel data transmission approach. Primarily, OFDM is more resilient to electromagnetic interference, and it enables more efficient use of total available bandwidth because the subchannels are closely spaced. It is also more resistant to interference because several channels are available.

Advanced error correction can be used to spread out the overall data and compensate for small errors. So, narrowband interference on a single subchannel will not affect the other channels, enabling the overall system to still operate. Frequency-selective interference fading due to multipath echo effects can also be corrected. The lower data rate on the individual subchannels enables guard intervals to be used between symbols, which eliminates

intersymbol interference and helps with multipath errors.

There are two primary disadvantages with OFDM compared to single-channel systems. OFDM systems must have closely tuned transmitters and receivers. This requires the timing on signal modulators and demodulators be closely matched and produced to tight tolerances. It also makes the system more sensitive to Doppler shift and, therefore, less effective for high-speed moving vehicles.

OFDM also has several advantages compared to standard frequency-division multiplexing. The radio frequency receiver is simpler in OFDM because the entire signal can be received in a single frequency selective filter and separated in software using a fast Fourier transform, while an FDM system requires a separate RF bandpass filter for each channel. It also has better overall bandwidth efficiency. There are some disadvantages in that the higher overall peak-to-average power (PAPR) ratio requires less efficient linear transmission circuitry.

OFDM applications

Orthogonal frequency-division multiplexing is used in many technologies, including the following:

- Digital radio, Digital Radio Mondiale, and digital audio broadcasting and satellite radio.
- Digital television standards, Digital Video Broadcasting-Terrestrial/Handheld (DVB-T/H), DVB-Cable 2

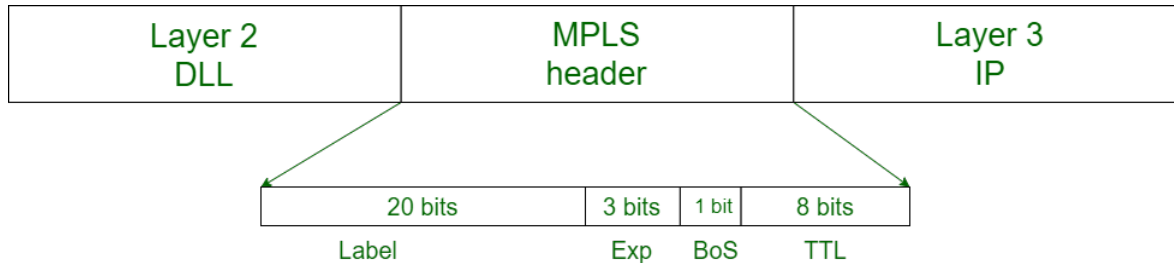
(DVB-C2). OFDM is not used in the current U.S. digital television Advanced Television Systems Committee standard, but it is used in the future 4K/8K-capable ATSC 3.0 standard.

- Wired data transmission, Asymmetric Digital Subscriber Line (ADSL), Institute of Electrical and Electronics Engineers (IEEE) 1901 powerline networking, cable internet providers. Fiber optic transmission may use either OFDM signals or several distinct frequencies as FDM.
- Wireless LAN (WLAN) data transmission. All Wi-Fi systems use OFDM, including IEEE 802.11a/b/g/n/ac/ax. The addition of OFDMA to the Wi-Fi 6/802.11ax standard enables more devices to use the same base station simultaneously. OFDM is also used in metropolitan area network (MAN) IEEE 802.16 Worldwide Interoperability for Microwave Access (WiMAX) installations.
- Cellular data. Long-Term Evolution (LTE) and 4G cellphone networks use OFDM. It is also an integral part of 5G NR cellular deployments.

□ Multi Protocol Label Switching (MPLS)

Multi Protocol Label Switching (MPLS) is an IP packet routing technique that routes IP packet through paths via labels instead of looking at complex routing tables of routers. This feature helps in increasing the delivery rate of IP packets.

MPLS uses layer 3 service i.e, Internet Protocol, and uses router as forwarding device. The traffic of different customers is separated from each other because MPLS works somewhat like VPN. It does not work like regular VPN that encrypts the data but it ensures packet from one customer cannot be received by another customer. An MPLS header is added to packet that lies between layers 2 and 3. Hence, it is also considered to be Layer 2.5 protocol.



MPLS Header :

The MPLS Header is 32 bit long and is divided into four parts :

- **Label** - This field is 20 bit long and can take value b/w 0 & 220 - 1.
- **Exp** - They are 3 bits long and used for Quality of Service(QoS).
- **Bottom of stack (S)** - It is of size 1 bit. MPLS labels are stacked one over other. If there is only one label remained in MPLS header, then its value is 1 otherwise 0.
- **Time to Live (TTL)** - It is 8 bit long and its value is decreased by one at each hop to prevent packet to get stuck in network.

Ktunotes.in

Multiprotocol Label Switching (MPLS) Routing

Forwarding in MPLS :

LSRs receive IP packet CE and add an MPLS header in between layer 3 and layer 2 means it encapsulates the link-layer i.e, layer 2 frames. This feature allows LSRs to support receiving packets containing frames from different protocols like Frame Relay, Metro Ethernet, etc, that's why it is called multi-protocol.

MPLS forwarding is based on label attached to IP packet. This label attachment is regulated by protocol called

Downloaded from Ktunotes.in

Label Distribution Protocol(LDP). Each LSR initially learns routes as normal routers do. This learning starts with PE routers. Each PE router learns routes to different subnets from CE router. Suppose PE router PE1 learns route to subnet (say subnet1) from CE router. Now PE1 will add label to packet, forward to its neighboring LSR, and tells them that if you receive packet which destination address to subnet1 then forward it to me.

Again this process is repeated by this LSR. In this way, LSR learns routes and add this information in Label Forwarding Information Base(LFIB). Now if any PE receives packet with destination to subnet1, then looking at labels and LFIB, LSRs can easily forward IP packet.

□ 10 pillars of 5G

Ktunotes.in

We have been witnessing an exponential growth in the amount of traffic carried through mobile networks.

According to the Cisco visual networking index, mobile data traffic has doubled during 2010-2011; extrapolating this trend for the rest of the decade shows that global mobile traffic will increase 1000x from 2010 to 2020.

The surge in mobile traffic is primarily driven by the proliferation of mobile devices and the accelerated adoption of data-hungry mobile devices especially smart phones

Ten pillars of 5G

1. Evolution of existing RATs
2. Hyperdense Small-Cell Deployment
3. Self- Organizing Network
4. Machine Type Communication
5. Developing Millimeter wave RATs.
6. Redesigning Backhaul links
7. Energy Efficiency
8. Allocation of New Spectrum for 5G
9. Spectrum Sharing
10. Ran Virtualization

1. Evolution of existing RATs

5G will hardly be a specific RAT, rather it is likely that it will be a collection of RATs including the evolution of the existing ones complemented with novel revolutionary designs.

The first and the most economical solution to address the 1000x capacity crunch is the improvement of the existing RATs in terms of SE, EE and latency.

Qualcomm have recently been working on developing LTE in the unlicensed spectrum as well as integrating 3G/4G/Wi-Fi transceivers into a single multi-mode base station (BS) unit.

2. Hyperdense Small-Cell Deployment

Hyperdense small-cell deployment is another promising solution to meet the 1000x capacity crunch, while bringing additional EE to the system as well.

☐ Large cell

- High power eNB
- Macro-eNB site can be difficult to find

☐ Small cell

- Low-power base station or RRH
- Hot-spot coverage
- Coverage in area not covered by the macro network
- Indoor coverage
- Small site size

In general, there are two different ways to realize HetNet:

- Overlaying a cellular system with small cells of the same technology, that is, with micro-, pico-, or femtocells;
- Overlaying with small cells of different technologies in contrast to just the cellular one (e.g. High Speed

Packet Access (HSPA), LTE, WiFi, and so on).

3. Self-Organizing Network (SON) As the population of the small cells increases, SON gains more momentum.

Almost 80% of the wireless traffic is generated indoors.

To carry this huge traffic, we need hyper dense small-cell deployments in homes installed and maintained mainly by the user - out of the control of the operators.

These indoor small cells need to be self-configurable and installed in a plug and play manner.

Furthermore, they need to have SON capability to intelligently adapt themselves to the neighboring small cells to minimize inter-cell interference.

4. Machine Type Communication (MTC) Machine type communication (MTC) is an emerging application where either one or both of the end users of the communication session involve machines. MTC imposes two main challenges on the network:

- The number of devices that need to be connected is tremendously large.
- The accelerating demand for real time and remote control of mobile devices (such as vehicles) through the

network

Ktunotes.in

5. Developing Millimeter-Wave RATs

Extremely high frequency EHF is the International Telecommunication Union (ITU) designation for the band of radio frequencies in the electromagnetic spectrum from 30 to 300 gigahertz(GHz).

Radio waves in this band have wavelengths from ten to one millimeter, giving it the name millimeter band or millimeter wave, sometimes abbreviated MMW or mmW. can be used for high-speed wireless broadband communications.

There are three main impediments for mmWave mobile communications.

- The path loss is relatively higher at these bands, compared to the conventional sub-3GHz bands.
- Electromagnetic waves tend to propagate in the Line-Of-Sight (LOS) direction, rendering the radio links vulnerable to being blocked by moving objects or people.
- The penetration loss through the buildings is substantially higher at these bands, blocking the outdoor RATs for the indoor users.

Advantage for mmWave communications

- An enormous amount of spectrum is available in mmWave band; for example, at 60 GHz, there is 9GHz of unlicensed spectrum available.

Ktunotes.in

- The small antenna sizes ($\hat{I} \gg \lambda/2$) and their small separations (also around $\hat{I} \gg \lambda/2$), enabling tens of antenna elements to be packed in just one square centimeter.

6. Redesigning Backhaul Links In parallel to improving the RAN, backhaul links also need to be reengineered to carry the tremendous amount of user traffic generated in the cells.

Otherwise, the backhaul links will soon become bottlenecks, threatening the proper operation of the whole system.

The problem gains more momentum as the population of small cells increases. Different communication mediums can

be considered, including optical fiber, microwave and mmWave.

7. Energy Efficiency (EE)

Today, Information and Communication Technology (ICT) consumes as much as 5% of the electricity produced around the globe and is responsible for approximately 2% of global greenhouse gas emissions roughly equivalent to the emissions created by the aviation industry.

What concerns more is the fact that if we do not take any measure to reduce the carbon emissions, the contribution is projected to double by 2020. Hence, it is necessary to pursue energy efficient design approaches from RAN and backhaul links to the UEs.

8. Allocation of New Spectrum for 5G

Another critical issue of 5G is the allocation of new spectrum to fuel wireless communications in the next decade. The 1000x traffic surge can hardly be managed by only improving the spectral efficiency or by hyper densification. In fact, the leading telecom companies such as Qualcomm and NSN believe that apart from technology innovations, 10 times more spectrum is needed to meet the demand.

9. Spectrum sharing

The efficient use of available spectrum is always of critical importance. Innovative spectrum allocation models

(different from the traditional licensed or unlicensed allocation) can be adopted to overcome the existing regulatory limitations. Plenty of radio spectrum has traditionally been allocated for military radars where the spectrum is not fully utilized all the time (24/7) or in the entire geographic region.

10. RAN virtualization

The last but not least critical enabler of 5G is the virtualization of the RAN, allowing sharing of wireless infrastructure among multiple operators. Network virtualization needs to be pushed from the wired core network (e.g. switches and routers) towards the RAN. For network virtualization, the intelligence needs to be taken out of the RAN hardware and controlled in a centralized manner using a software brain, which can be done in different network layers.

Virtualization can also serve to converge the wired and the wireless networks by jointly managing the whole network from a central orchestration unit, further enhancing the efficiency of the network. Finally, multi-mode RANs supporting 3G, 4G or WiFi can be adopted where different radio interfaces can be turned on or off through the central software control unit to improve the EE or the Quality of Experience (QoE) for the end users.